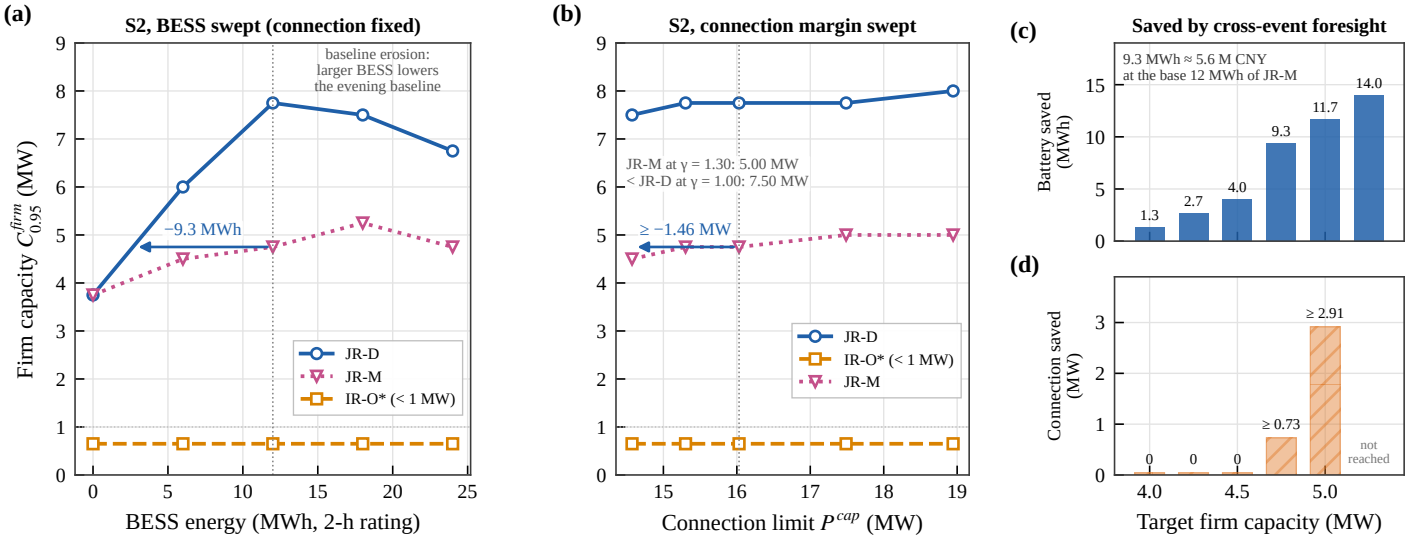
(a)
S2, BESS swept (connection fixed)
baseline erosion:
larger BESS lowers
the evening baseline
−9.3 MWh
JR-D
JR-M
IR-O* (< 1 MW)
Firm capacity $C_{0.95}^{firm}$ (MW)
BESS energy (MWh, 2-h rating)
(b)
S2, connection margin swept
JR-M at γ = 1.30: 5.00 MW
< JR-D at γ = 1.00: 7.50 MW
≥ −1.46 MW
JR-D
IR-O* (< 1 MW)
JR-M
Connection limit $P^{cap}$ (MW)
(c)
Saved by cross-event foresight
9.3 MWh ≈ 5.6 M CNY
at the base 12 MWh of JR-M
Battery saved (MWh)
1.3
2.7
4.0
9.3
11.7
14.0
(d)
Connection saved (MW)
0
0
0
≥ 0.73
≥ 2.91
not reached
Target firm capacity (MW)